

YALUTOROVSK, Russian Federation

WMO: 284650

Lat: 56.683N

Lon: 66.350E

Elev: 61

StdP: 100.59

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.7	-30.8	-36.7	0.1	-33.5	-33.7	0.2	-30.8	9.2	-10.5	8.4	-10.5	1.6	190	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	29.7	20.3	28.0	19.9	26.2	18.9	21.9	27.5	20.8	26.4	19.8	24.9	3.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.7	25.2	18.8	13.7	23.8	17.8	12.8	22.8	64.1	27.6	60.5	26.5	57.0	25.0	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.2	7.4	-36.5	32.7	3.4	1.8	-39.0	33.9	-40.9	35.0	-42.8	36.0	-45.3	37.3		
(5)			-36.1	23.2	3.7	1.4	-38.8	24.2	-41.0	25.0	-43.1	25.7	-45.8	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-17.0	-14.6	-5.9	4.2	12.1	17.2	18.4	16.4	10.4	3.2	-6.3	-13.5	(6)
(7)		DBStd	13.85	7.49	7.55	5.43	5.47	5.22	4.47	3.73	4.00	4.29	4.88	7.38	8.05	(7)
(8)		HDD10.0	3731	837	689	493	186	38	4	0	2	46	216	490	729	(8)
(9)		HDD18.3	6042	1096	923	752	423	203	74	47	88	240	469	740	988	(9)
(10)		CDD10.0	857	0	0	0	13	103	219	261	199	58	5	0	0	(10)
(11)		CDD18.3	127	0	0	0	0	9	38	50	27	2	0	0	0	(11)
(12)		CDH23.3	1204	0	0	0	4	113	393	443	232	18	0	0	0	(12)
(13)		CDH26.7	301	0	0	0	0	26	112	106	56	1	0	0	0	(13)
(14)	Wind	WSAvg	3.3	3.2	3.4	3.9	3.9	3.8	3.2	2.8	2.8	3.1	3.4	3.4	3.3	(14)
(15)	Precipitation	PrecAvg	448	20	14	19	19	41	63	76	54	47	35	33	26	(15)
(16)		PrecMax	621	41	30	43	45	102	176	141	90	95	91	69	54	(16)
(17)		PrecMin	312	1	1	1	0	6	12	18	24	9	0	7	10	(17)
(18)		PrecStd	66	11	7	11	12	23	33	34	21	21	22	14	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	2.8	7.8	23.4	29.6	31.4	31.5	30.6	25.6	18.5	7.7	2.4	(19)
(20)			MCWB	-0.1	1.2	4.2	13.0	16.4	20.3	21.7	20.7	17.1	11.5	5.5	0.9	(20)
(21)		2%	DB	-1.4	0.7	4.6	18.9	26.2	29.6	29.3	28.1	22.6	15.1	5.7	1.0	(21)
(22)			MCWB	-2.3	-0.5	2.1	11.1	15.8	20.0	21.3	20.3	15.8	9.9	4.5	0.0	(22)
(23)		5%	DB	-3.3	-1.3	2.9	15.6	23.8	27.7	27.6	25.8	19.7	12.4	3.6	-1.1	(23)
(24)			MCWB	-4.0	-2.2	0.9	9.6	15.0	19.3	20.6	19.4	14.2	8.5	2.6	-1.8	(24)
(25)		10%	DB	-6.3	-3.7	1.6	12.9	21.4	25.5	25.9	23.8	17.5	10.3	2.1	-3.2	(25)
(26)			MCWB	-6.9	-4.6	0.0	8.1	13.8	18.5	19.6	18.5	13.0	7.4	1.1	-4.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	1.6	4.7	13.8	19.1	22.9	23.6	22.4	18.5	12.5	6.1	1.1	(27)
(28)			MCD B	0.7	2.5	6.9	20.9	25.0	28.7	28.4	27.9	23.9	16.8	7.1	2.0	(28)
(29)		2%	WB	-2.2	-0.4	2.2	11.7	17.2	21.2	22.1	21.1	16.3	10.5	4.3	0.0	(29)
(30)			MCD B	-1.6	0.6	4.1	17.3	23.8	27.5	27.7	26.4	21.2	14.4	5.5	0.9	(30)
(31)		5%	WB	-3.9	-2.1	1.2	10.1	15.8	20.1	21.2	19.9	14.8	8.9	2.7	-1.8	(31)
(32)			MCD B	-3.2	-1.3	2.7	15.5	22.3	26.1	26.6	24.8	18.8	11.8	3.7	-1.2	(32)
(33)		10%	WB	-6.8	-4.5	0.2	8.3	14.3	19.0	20.2	18.8	13.5	7.5	1.1	-3.9	(33)
(34)			MCD B	-6.3	-3.8	1.5	12.6	20.4	24.5	25.1	23.2	16.8	10.1	1.9	-3.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	9.2	9.2	9.4	10.9	10.6	10.1	10.1	9.0	7.0	5.9	7.1	(35)
(36)			MCDBR	7.6	8.2	8.2	13.8	14.4	14.3	13.3	13.4	12.5	11.3	5.7	6.5	(36)
(37)			MCWBR	7.3	7.4	6.4	7.8	6.7	6.5	6.4	6.5	6.7	6.9	4.8	6.3	(37)
(38)		5% WB	MCDBR	7.5	8.0	7.7	12.7	13.0	12.4	12.2	12.2	11.6	10.7	5.3	6.4	(38)
(39)			MCWBR	7.3	7.4	6.3	7.6	6.8	6.3	6.7	6.5	6.6	7.0	4.7	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.314	0.371	0.414	0.411	0.409	0.423	0.432	0.386	0.354	0.312	0.284	(40)	
(41)		taud	2.033	2.027	1.944	2.087	2.223	2.299	2.256	2.198	2.287	2.240	2.112	1.997	(41)	
(42)		Ebn at Noon	562	707	755	792	826	834	810	770	753	665	523	440	(42)	
(43)		Edh at Noon	62	96	136	138	128	121	124	123	97	78	58	50	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.77	1.77	3.31	4.47	5.29	5.74	5.46	4.25	2.74	1.39	0.67	0.46	(44)	
(45)		RadStd	0.06	0.11	0.19	0.39	0.41	0.47	0.51	0.44	0.27	0.23	0.10	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page